# The Remaining Mathematical Obstruction in the Brenier Formalization

## Purpose

This note explains the exact mathematical issue in the current formalization of Brenier's theorem. The short version is:

- the repository already formalizes the chain

$$\text{convex potential} \Longrightarrow \text{subgradient support} \Longrightarrow \text{gradient map a.e.} \Longrightarrow \text{graph plan};$$

- the remaining gap is the passage

$$\text{optimal quadratic coupling} \Longrightarrow \text{convex potential},$$

  in the full classical generality;

- the extra hypothesis `HasRockafellarBound` is not a genuine Brenier assumption, but an artifact of the current real-valued Rockafellar construction.

## What the code currently proves

Let $\pi \in \Pi(\mu, \nu)$ be a coupling on a finite-dimensional real inner product space $E$.

The repository currently proves the following.

### 1. If one already has a convex potential, the plan collapses to a map

Suppose $\phi : E \to \mathbb{R}$ is convex and

$$\operatorname{supp}(\pi) \subseteq \operatorname{Graph}(\partial\phi).$$

If $\mu \ll m$, where $m$ is an additive Haar measure, then:

1. $\phi$ is differentiable $m$-almost everywhere;
2. hence $\phi$ is differentiable $\mu$-almost everywhere;
3. at points of differentiability,
$$\partial\phi(x) = \{\nabla\phi(x)\};$$
4. therefore $\pi$ is concentrated on the graph of $\nabla\phi$;
5. hence
$$\pi = (\mathrm{id}, \nabla\phi)_{\#}\mu, \qquad (\nabla\phi)_{\#}\mu = \nu.$$

This part is formalized.

### 2. If one already has a real-valued Rockafellar potential, the plan collapses to a map

The present Rockafellar file constructs a function

$$\phi_\Gamma(x) = \sup \mathcal{V}_\Gamma(x)$$

for a rooted family of finite-chain values $\mathcal{V}_\Gamma(x)$, but only under the hypothesis that this supremum is finite for every $x$.

This is encoded by the predicate `HasRockafellarBound`.

Under that hypothesis, the code proves:

1. $\phi_\Gamma : E \to \mathbb{R}$ is convex;
2. $\Gamma \subseteq \mathrm{Graph}(\partial\phi_\Gamma)$;
3. therefore, if $\Gamma = \mathrm{supp}(\pi)$ and $\mu \ll m$, then

$$\pi = (\mathrm{id}, \nabla\phi_\Gamma)_\# \mu.$$

### 3. Final packaged theorem currently available

The current endpoint is:

> If $\pi$ is an optimal quadratic coupling, $\mu \ll m$, and $\pi$ satisfies `HasRockafellarBound`, then there exists a convex $\phi$ such that
>
> $$\pi = (\mathrm{id}, \nabla\phi)_\# \mu, \qquad (\nabla\phi)_\# \mu = \nu,$$
>
> and the graph plan of $\nabla\phi$ is optimal.

## What `HasRockafellarBound` actually says

Fix a set $\Gamma \subseteq E \times E$ and a base point

$$(x_0, y_0) \in \Gamma.$$

The rooted Rockafellar value set at $x \in E$ is made of all finite-chain expressions

$$\sum_{i<n} \langle y_i, x_{i+1} - x_i \rangle + \langle y_n, x - x_n \rangle$$

over nonempty chains

$$[(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)] \subseteq \Gamma$$

starting at the chosen base point.

Then `HasRockafellarBound` says:

> there exists a base point in $\mathrm{supp}(\pi)$ such that for every $x$, the set of all those chain values is bounded above in $\mathbb{R}$.

Equivalently, the current code assumes that the pointwise supremum defining the Rockafellar potential is finite everywhere.

## Why this is not part of classical Brenier

In the standard mathematical proof of Brenier's theorem, one does *not* assume such a bound.

The reason is that one does not start by insisting that the Rockafellar potential be $\mathbb{R}$-valued everywhere.

Instead, one usually proceeds in one of the following equivalent ways:

1. define a proper convex potential that is allowed to take the value $+\infty$;
2. or define the potential on the convex hull of the relevant domain only;
3. or prove finiteness only on the set where it is later needed.

In all of these formulations, the supremum is allowed to exist without a global bounded-above hypothesis on every chain-value set.

So the mathematical issue is *not* that Brenier truly needs an extra assumption. The issue is that the present formalization took the convenient route

$$\phi_\Gamma : E \to \mathbb{R}$$

too early.

## Where the current proof becomes too rigid

The rigid point is the combination of the following two decisions:

1. define the Rockafellar potential as an actual real-valued supremum;
2. feed that directly into the real-valued convex and differentiability API.

That gives a clean downstream pipeline, but it forces an upstream finiteness hypothesis.

More precisely:

- the current Rockafellar theorem proves

  $$\Gamma \subseteq \mathrm{Graph}(\partial\phi)$$

  only after one knows $\phi : E \to \mathbb{R}$;
- but classical Rockafellar gives first a *proper convex* potential, not necessarily finite on all of $E$;
- the current `ConvexAE` file is a theorem about real-valued convex functions $\phi : E \to \mathbb{R}$, not proper extended-real convex functions.

So the obstruction is structural:

> the present upstream API is too narrow for the classical Rockafellar construction, while the present downstream API expects exactly that narrow real-valued input.

## What the missing theorem should conceptually look like

The ideal mathematical statement is something like:

> If $\Gamma \subseteq E \times E$ is cyclically monotone for the pairing, then there exists a proper convex function $\phi$ such that
>
> $$\Gamma \subseteq \mathrm{Graph}(\partial\phi).$$

Then, in the Brenier setting, one proves that $\phi$ is finite $\mu$-almost everywhere on the source side, and therefore that $\nabla\phi$ is meaningful $\mu$-almost everywhere.

That is the standard textbook route.

## Why the issue is genuinely mathematical, not merely technical

One might ask whether the current bound can simply be deduced from cyclical monotonicity.

In general, no: cyclical monotonicity alone does not force a globally finite real-valued Rockafellar supremum on all of $E$.

What cyclical monotonicity gives is the geometric structure needed for a supporting convex object. But that object may naturally be proper and extended-valued.

So the distinction is:

- *classical Rockafellar*: a proper convex potential;
- *current implementation*: a globally finite real-valued potential.

The latter is stronger, and that is exactly where the extra assumption enters.

## What a correct fix would look like

To remove the gap in the mathematically right way, the code should be refactored along the following lines.

### Option A. Extended-real Rockafellar first

1. Replace the current real-valued Rockafellar potential by a proper convex potential allowed to take the value $+\infty$.
2. Prove the Rockafellar containment theorem at that level:
$$\Gamma \subseteq \mathrm{Graph}(\partial\phi).$$
3. Then prove a Brenier-specific finiteness theorem showing that in the quadratic optimal transport setting, $\phi$ is finite $\mu$-almost everywhere.
4. Finally adapt the downstream gradient-collapse argument to that proper-convex setting.

### Option B. Localized real-valued potential

1. Restrict the construction to a set on which the supremum is known to be finite.
2. Prove that this suffices for the support of the optimal plan.
3. Run the differentiability and graph-plan argument only on that region.

This is more ad hoc, and less faithful to the standard convex-analytic statement, but may be easier to formalize.

## Why I stopped before coding this refactor immediately

The missing step is no longer a short theorem in the current language. It requires choosing a new mathematical interface.

In particular, one has to decide:

- what codomain to use for the proper convex potential;
- how to formulate subgradients for that codomain;
- how to connect that formulation back to the existing real-valued differentiability theorems.

This is why the current code isolates the issue but does not yet resolve it completely.

## The shortest conversation to have next

The right next design question is:

> Should the repository move to a proper extended-valued convex potential, or should it localize the real-valued construction only where the Brenier argument needs finiteness?

That is the genuine mathematical fork in the road.